

Financial Risk Analysis using Python

Project Overview

This project analyzes customer transaction data to identify financial risk patterns, customer behavior, and suspicious activities using Python-based data analysis techniques.

Objective

The objective of this project is to analyze financial transaction behavior and identify high-risk customer accounts through segmentation, anomaly detection, statistical testing, and visualization.

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

Dataset Features

The dataset contains:

- Customer ID
- Account ID
- Account Type
- Transaction Date
- Transaction Type
- Transaction Amount
- Account Balance
- Risk Score
- Credit Rating
- Tenure Months

Data Cleaning

- Removed missing and duplicate values
- Converted dates and numerical fields
- Standardized account categories
- Processed transaction information for consistency

Exploratory Data Analysis

Performed analysis on:

- Debit vs Credit transactions
- Customer activity trends
- Overdraft behavior
- High-risk customer patterns
- Balance volatility
- Suspicious transaction detection

Customer Segmentation

Customers were segmented into:

- High Activity
- Medium Activity
- Low Activity

Based on:

- Transaction frequency
- Balance behavior
- Transaction volume

Anomaly Detection

Applied:

- IQR Method
- Z-score Method

To detect:

- Suspicious transactions

- Large withdrawals
- Abnormal financial behavior

Hypothesis Testing

Performed statistical testing to evaluate whether high transaction volume accounts maintain higher average balances than low-volume accounts.

Visualizations

Created charts using Matplotlib and Seaborn:

- Line charts
- Bar charts
- Histograms
- Scatter plots
- Heatmaps

Key Findings

- Frequent overdrafts indicated higher financial risk.
- High transaction activity did not always imply stable balances.
- Suspicious transaction behavior was identified through anomaly detection.

Recommendations

- Monitor high-risk accounts regularly.
- Implement real-time fraud monitoring.
- Improve customer segmentation strategies.
- Use predictive analytics for future risk estimation.

Conclusion

This project demonstrates how Python-based financial analysis can help identify risk patterns, improve customer monitoring, and support data-driven business decisions.